CSEB428: SOFTWARE TESTING AND AUTOMATION

L: 3 T: 0 P: 2, Credit: 4

Course Outcomes: Through this course students should be able to

CO1: Describe fundamental testing principles and construct basic unit tests using JUnit and test coverage techniques.

CO2: Apply structured approaches to design test plans, analyse requirements, and document test reports and defects effectively.

CO3: Apply black-box and white-box testing strategies to design effective and meaningful test cases.

CO4: Use requirement-based technique to design test cases using behaviour-driven development tools to ensure testability and comprehensive coverage.

CO5: Apply automated testing techniques to test using static, dynamic, and mutation analysis.

CO6: Apply automation testing technique to design script for web and mobile applications.

Unit 1: Software Testing Foundations

Introduction to software testing, Software testing principles, Test case and test suite design, Types of testing (unit, integration, system), JUnit basics and assertions, Code coverage metrics, Mutation testing

Unit 2: Test Planning and Documentation

Role of testing in SDLC, Test planning and estimation, Requirement analysis for testing, Test report writing, Defect lifecycle and bug reporting, Risk-based testing, Test Doubles, Assessing Adequacy and Code Coverage Analysis with JaCoCo, Flakey Tests.

Unit 3: Black-box and White-box Testing Techniques

Introduction to Test Selection and Test Adequacy, what are Test Obligations, Factors Influencing Test Effectiveness, Program Structure: Reachability and Observability, Mutation Testing Revisited, Program Structure and Fault Finding, Test Oracles, Oracles and Fault Finding,
Partition Testing, Combinatorial Testing, Requirements Coverage

Unit 4: Requirement Based Testing

Writing Requirements for Testability, Bad Requirements for Testability, Writing Test Cases for Requirements, Introduction to User Stories and Behaviour-Driven Development, Cucumber and Gherkin, Cucumber and Gherkin: How Does It Work?, Regular Expressions and Testing with Cucumber, Installing Cucumber and Configuring Eclipse Project, Creating Gherkin Scenarios and Step Definitions, Using Data Tables, Cucumber and Code Coverage

Unit 5: Automated Testing

Introduction to Automated Analysis, Automated Analysis Techniques, Automated random test generation, Symbolic and static analysis, Program Wellformedness Properties, Static Analysis with Infer, Metaheuristic Search, Mutation analysis tools, Dynamic test generation, Automating

Regression Testing, Automating Security Testing Using Fuzz Testing, Using Multiple Methods Effectively, The Evolution of Software Testing

Unit 6: Selenium-Based Web & Mobile Tests

Introduction to Web and Mobile Testing, Introducing Selenium, Web Test Planning, Minimal Essential Test Strategy (METS), Selenium WebDriver basics, Locators and browser interaction, Security and OWASP Top Ten Risks, Introduction to Performance Testing, Introduction to JMeter, Automated mobile app testing, Test execution with Appium

Examination Category: 11 (Mid Term Examination: All MCQ and End Term Examination: All MCQ)

Weightage: Mid Term Examination 40% Marks and End Term Examination 60% Marks.

Continuous Assessment: Not Applicable (Only Mid Term and End Term exams will be conducted)

- **Eligibility Criteria for Mid Term and End Term Examinations**

To be eligible for the **Mid Term Examination**, a student **must successfully complete** and produce **Coursera completion certificate** for **01 course included in Units 1, 2, and 3 on or before 12th February 2026 and the courses are listed here:**

1. Introduction to Software Testing

- **Eligibility Criteria for End Term Examinations**

To be eligible for the **End Term Examination**, a student **must successfully complete** and produce **Coursera completion certificates** for **all 04 courses included in all six units on or before 18th April 2026 and the courses are listed here:**

1. Introduction to Software Testing
2. Black-box and White-box Testing
3. Introduction to Automated Analysis
4. Web and Mobile Testing with Selenium

Course Link: https://coursera.org/programs/d-cseb428-software-testing-and-automation-mmm8a